

ASPIRE COLLEGE VEHARI CAMPUS

Notes • BS CS 3rd Semester • Lecture 15: IPv4 Addressing & Subnetting

Course: Computer Networks (Credit Hours: 3)

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1. WHAT IS AN IPV4 ADDRESS?

An **IPv4 Address** is a 32-bit logical address assigned to every device connected to a network so it can be uniquely identified and located.

Format: Written as 4 decimal numbers separated by dots (Dotted Decimal Notation), e.g., 192.168.10.1. Each decimal number is called an **Octet** (8 bits). Total size = **4 Octets × 8 bits = 32 bits**.

2. CLASSFUL IP ADDRESSING ARCHITECTURE

Class	First Octet Range	Default Subnet Mask	Designed For
Class A	1 - 126	255.0.0.0 (/8)	Massive networks (very few networks, millions of hosts).
Class B	128 - 191	255.255.0.0 (/16)	Medium-to-large organizations and universities.
Class C	192 - 223	255.255.255.0 (/24)	Small local networks (up to 254 hosts per network).
Class D	224 - 239	No subnet mask	Reserved strictly for Multicasting .
Class E	240 - 254	No subnet mask	Reserved for experimental / R&D purposes.

Note: The address 127.0.0.1 is specially reserved worldwide for **Loopback Testing** (checking local NIC hardware).

3. STRUCTURE OF AN IP ADDRESS: NETWORK ID VS. HOST ID

Network ID (NetID)

Identifies the specific neighborhood/sub-network where the device lives.

- All devices on the same physical wire share the same Network ID.
- Routers inspect this part to steer traffic.

Host ID (HostID)

Identifies the individual workstation, laptop, or printer within that specific network.

- Must be unique inside that local network.
- Cannot be all 0s (reserved for Network Name) or all 1s (Broadcast Address).

4. WHAT IS SUBNETTING? (DIVIDING BIG NETWORKS)

Concept: Subnetting is the process of splitting a single large physical network into two or more smaller, manageable logical networks (called **Subnets**).

How it works: The network borrows host bits and converts them into subnet bits.

Key Benefits:

- Prevents IP address waste.
- Reduces network congestion by shrinking broadcast domains.
- Enhances institutional security by isolating lab departments (e.g., Accounts vs. Students).

5. QUICK STUDENT REVIEW QUESTIONS

Q1: Why can we only assign 254 usable host IPs in a standard Class C network (/24)?

Ans: Because 1 octet has 8 host bits ($2^8 = 256$ addresses). Address **0** is reserved as the Network ID, and **255** is reserved for Directed Broadcast, leaving exactly $256 - 2 = 254$ usable hosts.

Q2: What is the main purpose of a Subnet Mask?

Ans: It tells routers and computers exactly which bits belong to the Network ID and which bits belong to the Host ID.